

depression, hopelessness, loneliness, explicit suicidal thoughts, farewell expressions, or potential behavioral intent are assigned more risk-sensitive emojis, such as skulls, crying faces, warning signs, pills, holes, and broken hearts. This demonstrates that the generated emoji sequences are not limited to coarse-grained sentiment polarity, but can also capture fine-grained psychological risk cues and severity differences. Therefore, the case study suggests that emoji-based emotional augmentation provides an intuitive and discriminative supplementary signal for suicide risk assessment.

B Experimental Results and Analysis

B.1 Datasets and Evaluation Metrics

Datasets. We evaluate our model and baseline methods on three widely used datasets for the SRD task, namely Reddit [7], Weibo, SuicidEmoji, and BigData 2025 Cup datasets. In the following, we describe each dataset in detail.

- **Reddit dataset**¹. This dataset was collected by the Subreddit community [7], which consists of reddit posts of 500 users across 9 mental health and suicide related subreddits. Each post is independently labeled by professional psychologists according to a five-level ordinal scale derived from the Columbia-Suicide Severity Rating Scale (C-SSRS) [6], including Support (SU), Indication (IN), Ideation (ID), Behavior (BR), and Attempt (AT)². In our experiments, we pre-processed and split the dataset into training (400 samples), validation (50 samples), and test sets (50 samples), with the Class-wise distribution: 20% SU, 20% IN, 34% ID, 15% BR, and 9% AT.
- **Weibo dataset**³. This dataset was first collected by Cao *et al.* [1] from the popular Weibo platform. On the benefit of the expert’s supervision reliability, 3,677 ordinary users were the study sample, all of whom had explicitly mentioned suicidal thoughts, suicide plans, or self-harm behaviors at least five times on different days. As a control group, they selected users who had never posted any content related to suicidal ideation and labeled them as non-suicidal. we pre-processed the data using standard text normalization procedures and randomly split the dataset into training (800 samples), validation (100 samples), and test sets (100 samples), respectively. The resulting splits are balanced across the two classes, suicidal (labeled as 1) and non-suicidal (labeled as 0) samples, facilitating stable training and fair evaluation.
- **SuicidEmoji dataset**⁴. This is an emoji-centric suicide-related social media dataset constructed from large-scale Reddit corpora, including SuicideReddit [4] and Robin [2], Suicide-related posts were primarily sourced from the */Suicide-Watch* subreddit. After removing duplicate posts and retaining only posts containing emojis, the dataset comprises 25,051 emoji instances, including 2,329

¹<https://github.com/bryant03/Sina-Weibo-Dataset>

²In the C-SSRS, users are classified into five ascending risk levels: Support (SU) < Indication (IN) < Ideation (ID) < Behavior (BR) < Attempt (AT), reflecting increasing suicide risk severity.

³<https://github.com/bryant03/Sina-Weibo-Dataset>

⁴<https://github.com/TianlinZhang668/SuicidEmoji/tree/main>

suicide-related posts and 22,722 control posts. Although the dataset does not rely on manual annotation, prior validation and the large-scale nature of the corpus ensure reliable self-labeling quality. The dataset provides detailed statistics on emoji usage frequency, diversity, sentiment polarity, and co-occurrence patterns, supporting emoji-aware suicide ideation detection and emoji prediction tasks. In our experiments, we adopt the remove version of the dataset, where emojis are removed and the remaining text is used as the textual input.

- **BigData 2025 Cup Dataset**⁵. This dataset originates from the IEEE BigData 2025 Cup challenge titled "Suicide Risk Prediction on Social Media." For the competition, the organizers originally partitioned the data into a publicly available training set (7,383 samples) and a held-out test set (1,283 samples). The ground-truth labels for the test set were withheld to ensure blind evaluation for the final ranking. Each social media post within the dataset is annotated with a suicide risk level according to a four-level ordinal scale, defined as follows: Indicator (IN), Ideation (ID), Behavior (BR), and Attempt (AT). In our experiments, we further divided the original training set into a new training (5906 samples), validation (739 samples), and test sets (738 samples).

B.2 Model Overall Performance Result

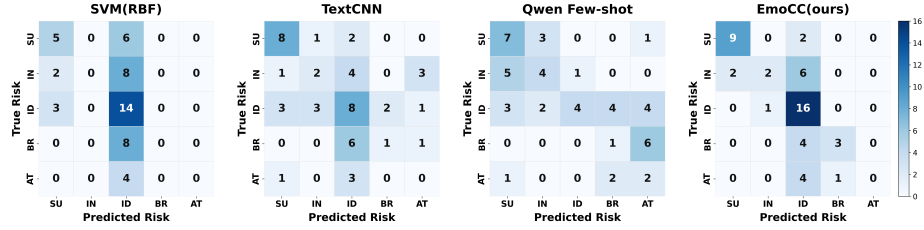


Figure 2: Normalized confusion matrix for prediction results on the Reddit dataset. A row in each figure represents examples of actual risk-level, the columns represent predicted risk levels. The values of diagonal elements represent the percentage of correct predictions.

B.3 Further Experiments

To evaluate the effectiveness of this transformation strategy, we conduct a comparative case study against representative emoji prediction models, including **DeepMoji** [3] and **EmojiLLM**[5], both of which are widely adopted for emotion-aware emoji generation.

when generating emojis for sentences that convey *distinct semantic meanings or nuanced emotional states*, the produced emoji outputs tend to exhibit overly high emotional similarity. Despite noticeable differences in linguistic expression or psychological intensity, the predicted emojis remain clustered within a narrow affective spectrum. This phenomenon weakens the intended emotion-enhancement effect, as the generated

⁵<http://www.bigdata2025.com/>

emoji signals fail to capture fine-grained emotional variations embedded in the original posts. We select an example from the Reddit dataset

The user’s post (custom_id="user-2_0): *Hi, so last night i was sitting on the ledge of my window contemplating whether or not i should jump. ... My dad had just choked me and told me that i should get out of the house, and earlier he had called me a bum. ... To me it seemed as though there was no reason to keep living, the people that are supposed to care about you the most dont. ...*

DeepMoji and **EmojiLLM** are applied in a direct inference manner (without any prompt-based constraint on emoji length or risk focus), whereas **Qwen-3-Max** is guided by our prompt to return a risk-aware sequence with **3–5 emojis**. We observe that DeepMoji and EmojiLLM tend to yield affectively generic or even positively biased emojis, which can obscure high-risk cues and reduce discriminative detail. In contrast, Qwen-3-Max produces a more risk-sensitive and context-consistent sequence that better reflects the post’s distress and suicidal intent, providing a stronger affective augmentation signal for downstream risk assessment.

C Case Study

C.1 Case Study

As illustrated in Fig. 3, we present a case study to analyze the post-level attention distribution across users with different suicide risk levels. Each column corresponds to a user categorized into Support (SU), Indication (IN), Ideation (ID), and Behavior (BR), where posts are arranged chronologically from past to present. Overall, the model effectively distinguishes varying levels of risk by adaptively assigning higher attention weights to posts that exhibit stronger negative emotions, suicidal intent, or behavioral signals, while lower-risk or supportive content receives relatively less attention. For the SU user, most posts convey positive and encouraging sentiments and are assigned low attention weights, indicating that the model correctly identifies them as non-risk signals. For the IN user, the model begins to focus on posts containing subtle distress or help-seeking expressions, showing sensitivity to early warning signals. For the ID user, posts explicitly expressing suicidal thoughts receive significantly closer attention, and the model also captures contextual cues such as distrust and social withdrawal, reflecting its ability to model deeper psychological states. For the BR user, the model primarily concentrates on recent posts that indicate severe conditions or imminent risk, assigning them the highest attention weights, which aligns with real-world risk assessment practices where recent behavior is more critical. In general, the attention distribution exhibits a clear temporal pattern, with more recent and high-risk posts contributing more to the final prediction. This demonstrates that the proposed post-level attention mechanism effectively captures both temporal dynamics and semantic severity, while emoji signals further provide complementary emotional cues that enhance the interpretability and robustness of the model.

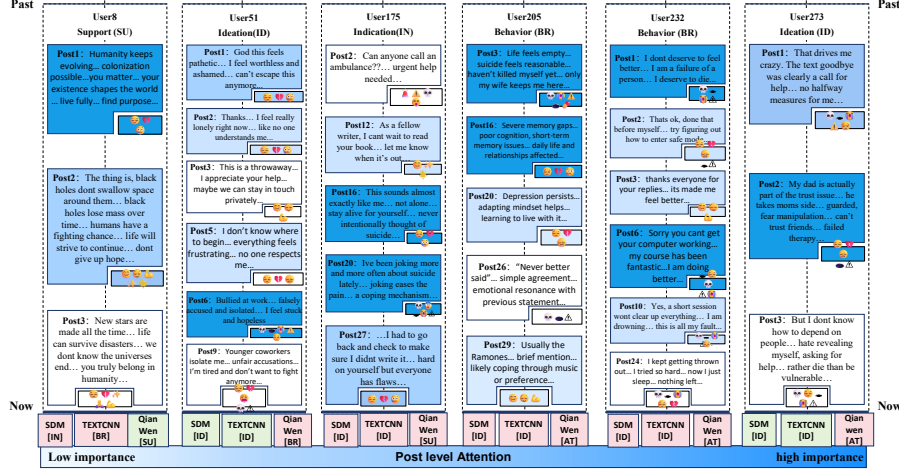


Figure 3: Case study of post-level attention across users with different suicide risk levels, augmented with prediction results from multiple models. Each column represents a user belonging to a specific risk category (SU, IN, ID, BR), where posts are arranged chronologically from past to present (top to bottom), highlighting the importance of the most recent (last) post. The color intensity (from white to deep blue) indicates the attention weight assigned by the model, reflecting the relative importance of each post in predicting the user’s risk level. At the bottom of each column, predictions from different models are presented, where green denotes correct predictions and red denotes incorrect predictions. This visualization demonstrates how the model emphasizes recent critical signals while revealing performance differences across models.

This case study highlights that conventional emoji prediction models tend to prioritize coarse-grained sentiment polarity (e.g., positive vs. negative) rather than fine-grained psychological expression, resulting in limited diversity and representational power. Such constraints reduce the effectiveness of emoji-based emotional augmentation. By leveraging more advanced large language models, our approach mitigates this issue and yields richer, more distinguishable emotion signals for subsequent risk assessment.

Table 1: Examples of posts and corresponding emoji labels.

Posts	Emoji sequence
1: 如果我不小心, 酒精真的会把事情搞砸。有些价值观和标准相近的朋友会让我觉得不那么孤独, 更外向、甚至有点自大; 但那些愚蠢的人(不得不聊天, 或被迫待在一群混蛋、保守派、“真信徒”、缺乏好奇心的人里)会让我烦躁, 最终愤怒, 把我从几乎持续的抑郁状态推向轻躁狂或混合状态。感谢氟哌啶醇, 它是我的“紧急刹车”, 防止我把一切搞砸。 Alcohol can seriously ruin things if I am not cautious. Certain friends with similar values make me feel less alone and more outgoing, but ignorant people trigger irritability and then anger, pushing me from a constant depressive state toward hypomania or a mixed state. Haldol is my emergency brake that keeps me from messing everything up.	🔥🚫🧠🙄🤔
2: 是的。一是我妻子, 二是我父母, 三是我曾经帮助过的一些青少年, 我把他们从自杀边缘拉了回来。如果我自杀了, 我知道这些人会感到难过, 而我可能会毁掉我为那些孩子所做的一切。所以, 本质上, 是内疚让我一天一天地假装活着。 Yes. My wife, my parents, and a few teenagers I have helped pull back from suicide. If I killed myself, they would be devastated, and I might undo the work I have done with those kids. Basically, guilt keeps me faking it day by day.	🧠🙄🤔🚫🔥
3: 感谢你的观察。生活一直很艰难。我只能短暂地把笑容贴在脸上、假装正常, 但我被责任、内疚与义务推着走。即使是在所谓“不错”的时候, 我也并不快乐; 当它结束时, 反而更觉得结束一切才合理。让我继续活着的唯一原因, 是不想让我的妻子失望。我的双相障碍对治疗不敏感, 服用了多种药物, 情况仍未好转。我不想再这样活下去。 Thank you for your observations. Life has been tough. I can fake a smile for a short time, but duty, guilt, and obligation keep pushing me forward. Even during so-called “good” times, I am not happy, and when they end it feels more reasonable to be done with everything. The only reason I am still alive is that I do not want to let my wife down. My bipolar disorder is treatment-resistant, and despite multiple medications, things do not improve. I do not want to keep living like this.	🧠🙄🤔🚫🔥
4: 可以试试 The Stooges 的歌曲: <i>Search and Destroy</i>、<i>I Want To Be Your Dog</i>、1969、<i>Gimme Danger</i>。 Give The Stooges a try: <i>Search and Destroy</i>, <i>I Want To Be Your Dog</i>, 1969, and <i>Gimme Danger</i>.	🧠🙄🤔🚫🔥
5: 这张图片断断续续当了我一年多的桌面背景。我觉得它非常准确地传达了我生活中那种忽起忽落的节奏和氛围。 This image has been my desktop background on and off for more than a year. I think it perfectly captures the stop-and-start rhythm of my life.	🧠🙄🤔🚫🔥
6: “唱我入睡, 唱我入睡, 然后让我一个人待着……不要在清晨叫醒我, 因为我将离去。”——The Smiths (我同意)。 “Sing me to sleep, then leave me alone... Don’t try to wake me in the morning, because I will be gone.” — The Smiths (I agree).	🧠🙄🤔🚫🔥

Table 2: Examples of posts and corresponding emoji labels from the BigData dataset.

Posts	Emoji sequence
帮帮我。有人能帮我在另一个国家找工作吗？也许这样能让我的人生走向更好的方向。 <i>Help. Can someone help me find a job in another country? Maybe it will turn my life in a good way.</i>	🙏🙏🙏🙏
今天过得很好，好到让我觉得这应该是人生的最后一天。 <i>Today was great, ready for it to be the last.</i>	😞💔😞
我几乎没有任何想要自杀的理由，但这种念头却一直挥之不去。 <i>I have almost no reason to want to kill myself, yet the thoughts are always there.</i>	😞💔😞
一切都结束了。我已经挣扎了很多年。我累了。 <i>It's over. I've been struggling for years. I'm tired.</i>	😞😞🙏
每年这个时候都是最难熬的。我讨厌我的生活。 <i>This time of year is the hardest. I hate my life.</i>	😞💔😞😞🙏
没有人愿意听我说话。我到底该怎么宣泄这些情绪？ <i>No one to listen. How am I supposed to vent all this?</i>	😞💔😞🙏🙏

Table 3: Examples of posts and corresponding emoji labels from the SuicidEmoji dataset.

Posts	Emoji sequence
我告诉了我暗恋的人我喜欢她，她的男朋友反而尊重我，因为我有勇气对一个已经有男朋友的人说出这件事，而且他们并不讨厌我。所以现在，我暗恋的人和她的男朋友反而成了我的朋友，也不觉得我是个多事的人，不会因为我没有把这件事憋在心里而嫌弃我。我真的很开心自己这样做了，把这些话说出来之后，我终于成功地放下并继续向前了。	😌😌👉👉👉👉
<i>i told my crush i liked them and her boyfriend respects me for being confident enough to say that to someone who already has a boyfriend and they don't hate me. so my crush and her boyfriend actually are my friends now and don't think i am a nuisance for not keeping it to myself. i'm so happy that i did this and i managed to finally move on by getting that off my chest.</i>	
去他妈的人生，我受够了。我不想再活下去了。我有性别焦虑（男跨女），可我永远也不可能成为真正的女性。我孤独、绝望、抑郁。我真的只想死。我甚至不知道自己为什么还要写这些，反正也没有人在乎。再见了。	💀😞😞😞👉👉💔
<i>fml, i'm done. i'm done living. i have gender dysphoria(mtf), i'll never be a female. i'm so lonely, hopeless, depressed. i really just wanna die. i don't know why i'm even writing this, no one cares anyways goodbye.</i>	
我真的非常想一走了之，可我还有责任在身。我得照顾我姐姐（那个“完美孩子”）的宠物，而她却在外地过着有生活、有朋友的人生。尽管我非常想说一句“去他妈的”，然后干脆一了百了，但我不愿向所有人证明他们是对的——证明我果然是个一无是处的废物。我甚至希望自己能被一辆卡车撞到。	💀😞😞😞👉👉💔
<i>i just want to go so badly but i have obligations. i'm supposed to be looking after my sisters (the golden child) pets while she's out of town having a life and friends. as much as i want to say fuck it and just get it over with, i don't want to prove everyone right that i'm a worthless fuck up after all. i hope i get hit by a truck</i>	
高期望。你好，我今年 14 岁，来自美国一个普通但非常勤奋的亚裔家庭。最近我被父亲不断地指责，直到我彻底崩溃，几乎做出了一些我真的不想谈及的事情。阻止我的是过去那些父母曾经关心我的时光、我新结识的朋友，以及给予我最美好童年的叔叔和阿姨。在我的成长过程中，我的大多数亲密朋友慢慢地搬走了，或者因为他们的改变而不再和我联系。我才十四岁，就已经在上高中，学业压力加上这些问题，真的让我承受不住。我的父母不理解我，在我抑郁的时候，只把我当成一个没用、没有希望的孩子。没有支持，没有爱，只有高得吓人的期望。请原谅我拙劣的表达。	😞😞😞😞👉👉😞
<i>high expectations hi i'm 14 and i come from a common hardworking asian family in the u.s. recently i have been criticized by my dad to the point where i broke down and almost attempted something that i really don't want to talk about. what stopped me is the past when my parents had time to care about me, my new friends, and my aunt and uncle that have given me the best childhood i had ever wanted. throughout my childhood slowly most of my close friends have moved away or stopped talking to me, because they have changed. i'm only fourteen and i attend high school, so homework along with these problems really got to me as a person. my parents don't understand me and only look at me as a hopeless child at the times of depression. no support, no love, just high expectations mind my use of writing</i>	
睡一觉就好了……就他妈睡一觉吧……至少你能安静、舒服几个小时…… <i>sleep it off...just fucking sleep it off..... at least u get some peace for a few hours</i>	😞😞😞😞

Table 4: Comparison of emoji sequences generation with different LLMs.

Model	LLM-Generated Emoji Sequence
Qwen-3-Max (prompt-constrained)	    
DeepMoji (direct output)	    
EmojiLLM (direct output)	     

References

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